

Zhihao QIAN

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EDUCATION

Xidian University

Sept. 2022 - Jul. 2026

BEng in Computer Science and Technology, Big Data and Intelligence Track

Xi'an, China

- GPA: 4/4
- Rank: 1/476 (second year), 1/109 (third year)

ACADEMICS EXPERIENCE

Chemical Reasoning with Reasoning-Oriented Large Language Models

Jun. 2025 - May 2026

Lead Researcher; initial research supervised by Prof. Shengchao Liu; undergraduate thesis advised by Prof. Yunan Li

- Engineered training pipelines for LLMs of various scales across diverse HPC environments, including Compute Canada (SLURM), Shanghai AI Lab (MegCompute), and AutoDL (On-Demand Instance).
- Achieved about 80% of the predictive performance of trillion-parameter LLMs with less than 0.5% of their model parameters and less than 5% of their token consumption.
- Explored explicit and latent chain-of-thought paradigms for chemical reasoning by reviewing LLM-based chemical reasoning, proposing latent-space mechanism dynamics, and completing the thesis with an Excellent grade.

LES-Talker: Fine-Grained Emotion Editing in Linear Emotion Space

Apr. 2024 - May 2025

Co-first Author (Accepted at IEEE Transactions on Affective Computing)

- Led the full research pipeline, from problem formulation to model design, experimentation, and paper writing.
- Proposed the Linear Emotion Space (LES), a novel interpretable framework enabling fine-grained emotion editing across types, intensities, and facial units.
- Designed LES-Talker with a universal Cross-Dimension Attention Network to align 3D model deformation with emotional control signals, achieving high-quality and controllable talking head generation.

Multimodal-MOF: Metal-Organic Framework Design

Oct. 2025 - Feb. 2026

Lead Researcher, Under the supervision of Prof. Shengchao Liu and Prof. Zhiling (Zach) Zheng

- Built a pretraining pipeline for multi-modal MOF-related data and ran initial experiments.
- Conducted exploratory trials toward downstream transfer, identifying key bottlenecks and refining the research direction based on preliminary findings.

KAN-Face: Efficient Resource Usage and Precision Lip-Sync

Mar. 2024 - Nov. 2024

Co-author (Published at ICASSP 2025)

- Contributed to the design of a lightweight framework utilizing KANs for efficient talking head generation.
- Actively contributed to rebuttal, focusing on highlighting contributions and addressing reviewer feedback.

EmoSpeaker: Emotion-Controlled Talking Face Generation

Nov. 2023 - Feb. 2025

Co-author (Accepted at IEEE Transactions on Multimedia)

- Assisted in a one-shot framework capable of fine-grained emotional control and precise lip synchronization.
- Contributed to revision and rebuttal writing, focusing on technical clarity and supplementary experiments.

Multi-modal Learning for Audio-driven Talking Head Generation

Nov. 2023 - Jan. 2024

Champion (Kaggle Competition), BDIV Lab

- As 1 of 2 core developers, achieved 4 #1 and 1 #2 ranks among 9 metrics in both video- and audio-based tracks.
- Modified SadTalker by removing PoseNet and redesigning ExpNet for direct 70D 3DMM prediction, enabling strategic trade-offs that improved key directional metrics.
- Enhanced realism via innovative blinking and lip-sync strategies using OpenFace and Hubert, improving non-English alignment and eye movement synthesis.

AWARDS

AAAI 2024 Workshop on AI for Digital Human (Task 3 Champion)

Comprehensive Development Scholarship (Second Prize in Sept. 2024)

Comprehensive Development Scholarship (First Prize in Sept. 2023)

DEVELOPER TOOLS

Proficient In: Python, PyTorch, C, SLURM, LaTeX

Used In Projects: MySQL, Assembly Language, C++, Qt Creator, MATLAB, Java